TECHNICAL ANNEX

September 22, 2025

1 EXCELLENCE IN S&T AND NETWORKING

A MAIN CHALLENGE

A.1 Description of Main Challenge

Europe has reached a critical moment in its transition to net zero and renewable energy. Under the Renewable Energy Directive EU/2023/2413, the EU committed to generating an ambitious 45% of its total energy from renewables by 2030, almost doubling its target since the original European Green Deal in 2019. The aim is to reduce greenhouse emissions, as well as to reach broader environmental goals, including its "Towards Zero Pollution for Air, Water and Soil" strategy. Offshore wind (OSW) has been identified as a key enabler of the renewable energy transition given stronger winds out at sea than onshore, reduced land use conflicts and visual impact over onshore wind. In particular, the North Sea, Baltic Sea, Celtic Sea, and parts of the Mediterranean offer ideal conditions for OSW development due to their consistent wind speeds, relatively shallow depths, and proximity to densely populated and industrialised regions [1, 2, 3, 4]. European governments (including the UK and Norway) awarded almost 20GW of new OSW energy in 2024, and yet current installed capacity stands at only 21GW (15GW of those in the UK), well short of the 158GW targeted for 2030. While it seems clear from this that a rapid expansion in OSW capacity is needed, such as expansion will bring significant challenges, including a risk of uneven socioeconomic benefits [5], heightened environmental and health risks [6, 7, 8], and a shortfall in coordinated training [1], safety planning [9], and workforce development [10] needed to support a resilient and inclusive energy transition. Besides Net Zero policies, a rapid expansion of European OSW is created by global competition and supply chain dependencies. Recent debates over the possible import of Chinese-manufactured turbines into European markets[1] illustrate the risks of technological dependence, effects on standards and safety oversight, and broader implications for politics, energy security and industrial competitiveness. These challenges underscore the urgent need for coordinated European action on OSW. While the Horizon programme offers generous funding for specific projects, these tend to deliver depth-first technological innovations within defined consortia. What they cannot provide is the same breadth of inter- and transdisciplinary networking, flexible knowledge exchange, and inclusive stakeholder engagement across countries and sectors that a COST Action enables. A networking grant complements Horizon's project-based focus by connecting otherwise fragmented initiatives, building durable research and policy communities, and creating the shared frameworks required to translate individual innovations into system-wide impact.

This COST Action explores the role that Artificial Intelligence (AI) and cyber-physical systems can play in streamlining and automating parts of the impact assessment process for offshore wind farms, logistics, and supply chain, including potential benefits, risks, knowledge gaps, and safety frameworks to enable rapid scaling-up of new projects while upholding commitments to nature, ocean health, and long-term sustainability. At the same time, it critically considers AI's own environmental footprint and the rising energy demands of data centres. Exploring innovative, energy-efficient approaches to AI is essential to ensure it supports (rather than undermines) the transition to renewables and the sustainable expansion of offshore wind. The SAFEWIND network will deliver knowledge generation and strategic roadmaps to tackle the following challenges associated with OSW development:

1 **Infrastructure and regulation** Despite growing political momentum, OSW installation rates are lagging behind projections, due to a number of identified factors (Wind Europe 2024), including requirements for grid and infrastructure upgrades, improved and streamlined planning and consenting processes and stronger engagement with local communities, all detailed in [11]. Firstly, the existing grid infrastructure in Europe has not expanded quickly enough to support the new OSW installations projected. This is due to technical complexity as well as regulatory delays. Secondly, port capacity is similarly limited with new facilities required to accommodate turbine assembly, transport, operations & maintenance activities.

[1] https://financialpost.com/pmn/business-pmn/octopus-energy-opens-door-to-bring-chinese-wind-turbines-to-uk

The current port infrastructure in Europe is expected to become insufficient to support the projected OSW expansion by 2029. With port expansions typically taking 6–10 years, this represents an urgent constraint. In a similar vein, vessel availability is currently limited to 80 offshore construction vessels, of which only five are able to support installation of the newer 14–15 MW turbines required to meet the 2030 targets. Deployment of related infrastructure, such as substations, are affected too. Finally, there is not currently a standardised European approach to permitting and planning new installations [12, 5], which means that many countries have experienced delays in consenting and planning, contributing to a mismatch between government targets and industry capacity to deliver. Recent advances in **AI can potentially help to automate consenting workflows**, using natural language processing to interpret regulatory texts, and applying predictive modelling to **forecast infrastructure bottlenecks**, including grid capacity, port readiness, and vessel availability. Machine learning can be used to **optimise logistics** for turbine assembly and maintenance [13], while digital twins and spatial modelling tools can **assess cumulative impacts and identify suitable installation zones** [14]. AI-driven stakeholder mapping and sentiment analysis [15] can also improve public engagement and reduce delays in planning.

2 **Environmental interaction and co-existence** At the same time as addressing political and logistic hurdles, an expansion of OSW means that its interaction with the marine environment (species, habitats, marine protected areas) and other maritime activities (fisheries, tourism, shipping, etc) becomes more pronounced. This can cause conflict with other sea users, such as fisheries, commercial shipping, oil and gas, aquaculture and birds, as well as tourism, giving rise to spatial conflicts and environmental trade-offs. To navigate such challenges, the EU's Marine Strategy Framework Directive (2008/56/EC) was adopted to coordinate the sustainable use of marine space across member states. This framework aims to facilitate coexistence between sectors, particularly in congested sea basins, by promoting transparent and environmentally responsible spatial allocation and mitigating user conflicts. However, while Marine Spatial Planning (MSP) is conceptually positioned as a solution to growing spatial tensions, its implementation is frequently hampered by regulatory inconsistencies, socio-economic pressures, and a lack of technical integration across jurisdictions [16, 17]. Particular uncertainty exists around OSW's ecological footprint, with many long-term impacts on environment and ecosystems still unknown [7, 18]. The EU's Marine Strategy Framework Directive (2008/56/EC) requires that member states achieve Good Environmental Status (GES) in their marine waters, ensuring that marine ecosystems remain healthy and resilient. Similarly, the EU Biodiversity Strategy for 2030, together with Natura 2000 obligations, requires member states to protect sensitive species and habitats even as they scale renewable infrastructure. To navigate this balance, environmental assessment frameworks such as the Environmental Impact Assessment (EIA) Directive (initially 85/337/EEC), the Strategic Environmental Assessment (SEA) and the Appropriate Assessment (AA) (for projects or plans that may affect the EU Natura 2000 network of protected areas) play a central role. The EIA, SEA and AA processes sets minimum requirements to assess potential ecological impacts with the aim of safeguarding biodiversity and mitigating climate change. However, while widely used, research has shown that these environmental assessments provide only broad guidelines and leave substantial discretion to national authorities in terms of the precise methodologies and indicators to be applied. Further complicating matters, the environmental licensing frameworks for renewable energy projects differ significantly across EU Member States, creating inconsistencies in how environmental standards are enforced [16]. While multiple studies have researched the impacts of OSW farms at different stages of their lifecycle, research has increasingly drawn attention to the multi-faceted ecological effects of OSW [19]. While short-term impacts are better understood, up to 86% of long-term effects remain uncertain, particularly for decommissioning phases, which most European OSW farms have not yet entered [20, 21]. **AI can enhance marine spatial planning and environmental assessment** by integrating large-scale, heterogeneous datasets, e.g. vessel traffic, biodiversity records, and oceanographic data, to model cumulative impacts and optimise site selection. Machine learning can identify spatial and temporal patterns of ecological sensitivity, predict species migration or disturbance zones, and **simulate multi-user scenarios** to minimise conflicts with fisheries, conservation zones, or shipping routes. AI can further **automate parts of the Environmental Impact Assessment process**, standardising indicators, detecting anomalies in monitoring data, and providing decision support under uncertainty—especially for long-term and decommissioning impacts where empirical data is lacking.

3 **Socioeconomic factors of growing OSW** The acceleration of OSW projects creates new socioeconomic opportunities and challenges, particularly for coastal communities. In terms of opportunities, communities may benefit from new investment, green job creation and infrastructure upgrades, which is relevant particularly to communities affected by a decline of traditional sectors such as fisheries, shipbuilding, and tourism [22]. However, OSW can also negatively affect the livelihoods of local communities due to its effects on fisheries and tourism through impacts on fishery stocks, ecosystem services and landscape/seascape. There is growing recognition that local voices should be incorporated into project planning, particularly during EIA procedures, to ensure that the distributional impacts of OSW are fairly as-

sessed and mitigated. Despite this, empirical evidence on the socioeconomic benefits remains limited. A meta-analysis by [23] found insufficient data to confirm that OSW reliably stimulates local employment or drives sustained social transformation. While initiatives like the Hornsea project in the UK have demonstrated positive collaboration with local authorities to promote training and supply chain development, such examples remain relatively isolated. Also, coastal communities often feel alienated from large-scale infrastructure projects, such as OSW, leading to opposition. The expansion of OSW also raises concerns regarding occupational health and safety (H&S). Available data suggest that OSW operations are three to four times riskier than those in other industrial sectors [9] and there is not currently a standardised EU-wide reporting framework for incidents. Risk exposure is exacerbated by remote locations, hazardous weather conditions, use of subcontractors amid supply chain pressures [10] as well as other accident risks such as ship collisions, meteorological hazards, etc. Human factors are reported to account for 60-90% of workplace accidents in OSW, with behavioural, psychological, and organisational dimensions all contributing [24]. Furthermore, the increasing automation of OSW operations introduces new human-machine interface risks, underlining the importance of training in both core technical competencies and emerging fields such as AI and cybersecurity. Addressing these comprehensively will be essential as the workforce is expected to grow substantially, with forecasts indicating 350,000 technician transfers annually in the near future [9]. An important task is to understand the full spectrum of risks and their origins in relation to OSW: ship collisions, technical issues, blade failures, and personal accidents all have a human as well as an environmental footprint, creating **new dynamic risk interactions** that require mapping and analysis. In sum, while OSW presents clear potential for green jobs and coastal renewal, it also raises challenges around risk, infrastructure, and equitable development. Delivering on its promise will require integrated planning, inclusive impact assessment, and targeted investment in workforce education and safety, linked with national education programmes, such as apprenticeships, university courses and training. **AI can play a key role in supporting the safe, inclusive growth of the OSW sector**. It can **enhance socioeconomic impact assessments** by combining spatial, demographic, and economic data to identify uneven outcomes and inform more equitable planning. AI-enabled models can **support public engagement** by analysing consultation responses and community sentiment, helping integrate local perspectives into decision-making. In workforce development, intelligent tutoring systems and adaptive learning platforms can **personalise and scale training**, aligning content with market needs and emerging skills in automation, AI, and cybersecurity. AI can also **improve occupational safety** by analysing sensor and incident data to predict risks, support safer task design, and monitor subcontractor performance, particularly as operations become more automated and dispersed.

The rise of AI, automation and cyber risks Artificial intelligence is rapidly transforming industries, including renewable energy and offshore wind, where it has been used in site planning [14], power and impact estimation [25, 26], techno-economic benefits forecasting [27], fault identification and [28] alarm code interpretation [29], structural health monitoring [30, 31], remaining useful life estimation of turbine components [32, 33], or for offering conversational decision support for maintenance and repair planning [34, 35]. This COST Action postulates that beyond these existing use cases, AI has significant potential in helping to advance OSW development and addressing persistent challenges around environmental and socioeconomic impact assessment, regulatory complexity, and workforce readiness. AI-powered modelling and remote sensing can enhance the precision and efficiency of ecological impact predictions, such as effects on bird migration, fish populations, and benthic habitats. In regulatory planning, AI tools can support marine spatial planning by integrating diverse datasets to identify optimal siting, reduce stakeholder conflict, and streamline licensing. AI-driven training platforms, simulations, and predictive safety analytics can help equip a growing workforce with the skills needed for increasingly automated and complex OSW operations, while improving health and safety outcomes.

However, there is also a need to understand the risks of AI, especially around the black box nature of leading AI frameworks and associated lack of transparency, which can obscure the basis for critical decisions in environmental assessment or operational control. Similarly, previous research in offshore wind has shown a lack of high-quality open datasets for model training and evaluation [36, 37]. This creates a risk of data imbalances and statistical biases leading to skewed or unreliable outputs, especially when datasets used for model training lack ecological or socioeconomic diversity, have sparse inclusion of infrequent weather events, feature an unbalanced representation of specific operational conditions [33], or cater inadequately for changing patterns due to climate change [38, 39, 40, 41, 42, 43]. As a result, AI models can collapse or become instable when exposed to poorly generalised data. In parallel, the sector faces growing cybersecurity vulnerabilities, particularly as AI systems become more integrated into core operational infrastructure [44, 45] and through increased automation [46, 47] such as in shipping, drones or maintenance robots. This risk is exacerbated by recent findings in the field of agentic AI, particularly involving large language models that perform tasks on behalf of users, which show that such agents can be easily jailbroken or manipulated to bypass safety constraints, enabling them to pursue harmful, illegal, or otherwise undesirable objectives [48]. Researchers have started to develop benchmarks and test suites to identify such behaviour [49, 50] as well as new strategies to enforce safely rails [51, 52, 53]. Yet with the rapid development in AI models, this field of research is still in its infancy, and rushed automation can represent a risk to infrastructure, operations and safety in OSW. Last but not least, there is a growing concern about the energy consumption and environmental impact of AI itself [54, 55, 56]. This brings an urgent need to investigate how AI-driven technology can be used to support renewable energy effectively and energy-efficiently without trading off climate goals, and meeting key performance indicators on energy and sustainability as set out e.g. in the Energy Efficiency Directive of 2024. Technological advances are needed on low-cost, efficient AI algorithms that decarbonise Europe's energy sector through accurate forecasting of wind generation output, real-time optimisation of energy dispatch and storage, coordination between generation, demand response, and storage, overall reducing the need for fossil fuel-based balancing power.

The role of climate change Besides AI, this COST Action considers the cross-cutting role of climate change and the high degrees of uncertainty inherent in all OSW-related tasks. Current climate policies are projected to result in warming of 2.7°C by the end of the century, compared to pre-industrial times; which pledges and targets as expected to reduce to 2.1°C, still higher than the Paris Agreement goal of 1.5°C[2]. There is high uncertainty about which future climate scenario will materialise, while within each scenario the spread from the individual climate models often exceeds the differences among scenarios. All the above imply potential changes in the wind climate, not only in terms of intensity, but also directions and previous research has shown that errors in wind estimates (e.g., 10%) can result in proportionally larger errors in wave height, often around 20% or more, particularly in wind-sea dominated conditions [57]. This creates knock-on effects for structural health monitoring [58], remaining useful life estimation, as well as vessel accessibility and operations & maintenance [59, 43]. The latter can be further exacerbated by sea level rise driven from global warming which is expected to alter several aspects of the continental shelf; e.g. bathymetry, shoreline position, waves and ocean circulation, among others. All the above further create uncertainties on future energy yield and resource estimation [42], the frequency and effect of extreme weather events [60, 41], which lead to different technology, turbine and coastal engineering requirements.

A.2 Relevance and Timeliness

This COST Action is both timely and urgent. With European OSW capacity currently sitting at 21GW, an almost **seven-fold increase is needed to reach the 2030 target** of 158GW and achieve corresponding climate and net zero goals laid out in European legislation, such as the Renewable Energy Directive EU/2023/2413. The next five years will therefore determine whether OSW remains a promise or becomes a cornerstone of Europe's clean energy future. Fast, transparent, and ecologically sound planning is essential to avoid trading one environmental crisis for another, as laid out in the EU's Biodiversity Strategy for 2030. While AI has significant potential to support the OSW sector in scaling up, this Action addresses the urgent need to ensure that safety risks at the intersection of AI and OSW are understood, research and evidence gaps identified, and mitigation strategies developed proactively. **Immediate areas of risk** include **transparency** of decision making by AI systems, risks to **safe operations** from **premature autonomy** of AI systems, **cyber and data integrity** threats, **uncertainty from climate** models and impacts of OSW on **marine species and habitats** and areas of ecological value such as Marine Protected Areas (MPAs). A clear regulatory and research agenda is needed to ensure that AI tools contribute to, rather than undermine, the safety, resilience, and legitimacy of offshore wind expansion, supporting inclusive growth. In essence, while it is clear that action is needed now and risks pertain, there is currently insufficient knowledge to define credible research priorities, establish clear policy directives, or develop robust standards that ensure AI contributes safely and equitably to the offshore wind transition.

B OBJECTIVES

This COST Action aims to build a balanced, inclusive, and geographically diverse pan-European network of experts to determine pathways towards a safe and sustainable digital-enabled growth of the OSW sector, enabled by novel advanced in AI. By identifying gaps in data, research, regulation and best practices, we will highlight solutions that align renewable energy expansion with social responsibility, environmental protection, and operational resilience. We will create roadmaps for:

(a) Safely automating parts of the environmental and socioeconomic impact assessment process via digitalisation, aligning with industry best practices, international regulations, and national laws, while protecting marine ecosystems and biodiversity towards long-term ocean health.

[2](Climate Action Tracker (2024). The CAT Thermometer. November 2024. Available at: https://climateactiontracker.org/global/cat-thermometer/ Copyright © 2024 by Climate Analytics and NewClimate Institute. All rights reserved, n.d.)

(b) Guiding the development and acceleration of new OSW projects and advancing new technology, such as floating, while maximising benefits and minimising environmental risks, drawing on insights from reliable public data and a digital environment.

(c) Enhancing workforce training and occupational safety by integrating digital tools and AI to reduce accident risks such as ship collisions or technical failures, while supporting education and upskilling to prepare a resilient, inclusive and diverse OSW workforce.

(d) Translating cutting-edge research into real-world practice through stakeholder collaboration, ensuring that advances in digital technologies and AI inform and enhance the OSW sector in the short and long term, while including the perspectives of and generating benefits for local and coastal communities.

As highlighted, there is a strong expectation that a rapid expansion of OSW will go hand-in-hand with increased automation in the sector driven by recent advances in generative AI, Large Language Models and the return of AI agents. This COST Action will work towards documenting the current state of the art in research and innovation at the intersection of OSW and AI, will identify critical research, data and knowledge gaps, and design road maps and training needs to inform future research and innovation in the OSW sector. To achieve this we define a set of research coordination and capacity building objectives below.

B.1 Research Coordination Objectives

This Action will collaborate through working groups to achieve the following key measurable objectives.

1. **Document, scope and define** the current and future **role of AI and digital technologies**, including hydro-physical simulation models, earth observation, and remote sensing, in automating geophysical and environmental profiling across European regions to **identify risks and opportunities of offshore projects to environment** (pollution, noise, habitat disruption), **commercial shipping** (including navigation safety, traffic flows, and interference with underwater cables and anchoring), **fisheries** (access restrictions, changing fish populations), **marine wildlife**[3], including birds, mammals and benthic species, as well as **coastal communities**, documented in open-access deliverables.

2. **Assess the relevance and accuracy of climate models** for understanding and forecasting the long-term impacts of extreme weather events and volatile sea conditions on offshore structures, ports and infrastructure, including improved predictive capabilities for storm surges, wave loading, sea-level rise, and shifting wind patterns that directly affect the durability, reliability, and maintenance of OSW assets, as well as resulting links with environmental and human health and safety impacts.

3. **Establish the effectiveness of digital twins and cybersecurity frameworks** in exposing vulnerabilities to physical and digital infrastructure towards the safe deployment of AI, automation and remote operations & maintenance in offshore wind farms; and **develop** robust, EU-aligned cybersecurity **protocols and risk governance strategies** to protect assets, personnel, and sensitive data.

4. **Create road maps** for the safe use of **large language models for multilingual information retrieval and synthesis to gather reliable evidence** from environmental reports, scientific articles and government documents to inform potential safety risks, benefits, and the effectiveness of different mitigation strategies, to ultimately bridge gaps between local knowledge, regulatory frameworks, and academic research, improving evidence-based decision-making that takes into account all stakeholders, i.e. from OSW owners/operators to local communities.

5. **Map human factors and scope inter- and trans-disciplinary training needs** in: (i) the offshore wind sector (safety, operations, environmental impact management) under challenging sea conditions, (ii) the digitalisation sector (skills in AI-driven automation, data interpretation, geospatial analysis), and (iii) risk management (hazard assessment, emergency response, cybersecurity awareness, regulatory compliance in increasingly automated and data-intensive environments).

These objectives will feed into working groups around the themes of **site planning**, **life cycle**, **technology**, **training** and **co-existence**, all under an over-arching and cross-cutting theme of **AI and digitalisation**.

[3] With a focus on areas of particular environmental value such as Marine Protected Areas (e.g. Natura 2000 sites) and other areas such as Important Areas for the Conservation of Marine Mammals (IMMAs), Sharks and Rays (ISRAs) or Birds (IBAs).

B.2 Capacity Building Objectives

- **Build a multi-, inter- and trans-disciplinary pan-European network of 100+ experts** who bring complementary perspectives on OSW that enables knowledge exchange and cross-border networking amongst a diverse set of actors and organisation types. The network will act as a knowledge and consultancy resource for policymakers, industry, and civil society, overcoming disciplinary silos to build joint, holistic perspectives on AI-enabled OSW development and fostering coordinated expertise across environmental, engineering, technical, and socioeconomic domains.
- **Create new cross-disciplinary education and research collaboration networks** across Europe through mobility programmes, joint training schools, and Short-Term Scientific Missions (STSMs). This will involve facilitating access to research infrastructure, labs and bespoke facilities, creating new and regular placements and internships to support doctoral and early-career researchers (ECRs) and innovators, and ultimately establishing a strong talent pipeline towards the 300K+ workforce required for the European OSW sector.
- **Support the formation of well-connected research clusters and consortia** that engage in joint collaborative research, and are able to rapidly respond to European and national funding opportunities. As part of this, we will promote collaborative doctoral training and co-supervision as well as cross-sector mentoring to develop future OSW and AI leaders that can drive innovation and build long-term partnerships.
- **Proactively address under-representation and intersectionality in the OSW sector** by promoting gender balance at 50%, geographical diversity including ultimately the majority of all 27 EU members states as well as affiliated countries, such as the UK as leaders in OSW, and the inclusion of under-represented groups across all career stages exceeding the OSW sector target of 33% [61]. This is to ensure equitable access to participation in training, leadership, and mobility opportunities, especially for researchers from ITC countries, early-career professionals, and those from minority backgrounds.
- **Develop active dissemination channels for diverse audiences**, including strategies for public engagement and community outreach, policy influence and academic visibility. Activities will include press releases, school visits, local community events such as information and discussion groups for community consultation, including coastal communities, Bachelor and Master student projects and industry placements to connect new talent with the OSW sector; open-access publications in scientific journals and authoring of white papers to capture the findings and outcomes of this Action, inform future science and encourage inclusive policy dialogue.
- **Collate, curate and release new datasets** that capture insights generated from this Action and that can enable informed decision making on future development, science and technology, as well as support research into OSW automation. The lack of representative, high-quality data is continuously raised as a barrier to progress in AI-driven OSW research [36, 37], which this Action will seek to address.

This Action has the collaboration and involvement of xxx partners from across Europe, including OSW leaders, such as the UK, Denmark and Norway, as well as countries with high potential in developing a strong OSW sector, but are still at the planning stage. A multi-actor approach is core to this collaboration to ensure that a diverse and comprehensive set of perspectives is represented. As such, we involve XX academic partners from across XX countries, balancing X and ICT countries, spanning disciplines such as economics, political science, social sciences, law, computer science, engineering, materials science, biology and earth sciences. We also involve XX industry partners (XX large enterprises and XX SMEs), and YY public sector partners, including government agencies, regulators, and advisory bodies. The initiative actively engages with a broad range of actors, including researchers, spatial planners, landowners, fisheries, local communities, professional associations, NGOs, civil society organisations, OSW operators and owners, and other marine users, to ensure inclusive, evidence-based networking and collaboration to inform sustainable sector growth.

C STATE-OF-THE-ART

State of the art research is discussed according to five themes: **site planning**, **life cycle**, **technology**, **training** and **co-existence**, which will inform the working groups for this Action in Section 3. All topics are considered under the cross-cutting theme of **AI and digitalisation** of the OSW sector.

C.1 Description of State of the Art

Site planning and Co-Existence Expanding OSW energy comes with the need to create new and more space for offshore wind farms, creating conflicts over the use of the marine environment with existing users, including fisheries, aquaculture, ship traffic, other renewable energy installations, oil and gas platforms, tourism, and others. Also, new sites are likely to coincide with existing and planned marine protected areas, creating conflict with conversation goals [62]. The EU adopted the new Biodiversity Strategy 2030 as part of the European Green Deal, which requires at least 30% of the marine area to be protected, yet leaves unspecified how this is to be implemented. In this regard, Puts et al. [17] show particularly that fencing off a larger area for biodiversity conversation will be more beneficial than multiple smaller areas as it allows ecological connectivity for animals to move freely within an area. These **competing use cases for limited space** have given rise to calls for a framework that can catalogue and weigh up stakeholder claims, benefits and costs in a transparent and uniform way [8]. For example, De Vasconcelos et al. [63] propose a standardised "one-stop-shop" procedure for new developments from a comparative study of Denmark, the UK and Taiwan, given that environmental licensing for renewable energy sources tends to specific to each country. A number of studies have proposed **algorithms to automate such space allocation constraints** in the identification of new OSW sites. For example, Virtanen et al. [2] apply a spatial prioritisation algorithm which balances social, ecological and economic factors against positive yield and least disturbance. Other research uses multi-criteria approaches to incorporate common exclusion criteria, including military areas, hydrocarbons and minerals, sand and gravel, aquaculture and fishing, marine renewable energy pilot zones, marine protected areas (MPAs), underwater lines and pipelines, maritime traffic, heritage areas, wind velocity, water depth, distance from shore [64]. A similar approach by Guşatu et al. [65] models risks and opportunities from the interaction between OSW farms and shipping, marine protected areas, fisheries and military activities. While impacts of multi-use scenarios have been studied, **climate change creates uncertainty** around future OSW yield and development sites, especially given **extreme weather conditions** [60, 41]. Martinez et al. [38] compare recent climate models under low, mid and high emissions scenarios and find reduced wind capacity in the North Atlantic, good conditions off the west coast of Ireland and limited potential in the Baltic Sea. Other studies report more fine-grained regional wind changes [66, 67], and discuss potential risks for structural health of turbines [58], yet overall acknowledge uncertainty depending on climate scenarios [42]. Rapella et al. [59] show that extreme wind events (extremely low or high winds), which do not allow OSW farms to operate, have become more frequent over time in a period of 1950-2020. Finally, **wake effects** are modelled when planning yield from wind farms [68], where some research demonstrates that a single OSW farm can reduce wind capacity by 20% for a span of 35-40km [69].

displacement of fishers to other areas, because the areas occupied by OSW become often inoperational for fishing, particularly for floating technologies

A number of papers focus on the **impacts of OSW on fisheries**, in terms of livelihoods as well as reducd catch and displacement as their original areas of operation become unavailable for fishing, particularly for floating technologies. Chaji and Werner [70] focus on economic tradeoffs of OSW considering fuel expenditures, insurance costs, fishing industry revenues, including income, livelihoods and fishing support businesses. The authors point to a lack of economic data or data collection efforts, lack of peer-reviewed methods to quantify economic impacts and a general lack of best practices in quantifying the full economic impacts of OSW on fisheries. Similarly, Bonsu et al. [12] discuss the socio-economic consequences of displaced fisheries, as well as ecological habitat transformation that happens as part of OSW installations. They point to a general gap and lack of scientific evidence to support better decision making in this area, specifically around the bottom-fixed and floating structures. While some papers take a stance of competing use [17] as above, others have focuses on **multi-use scenarios** [16], which have been shown to be viable, e.g. for trap and pot fisheries for crustaceans [12, 19] and are welcomed by fisheries [16] and anglers [71], though not always supported by OSW operators [72]. Similar concerns affect tourism through the effects of OSW on seascape, landscape and ecosystem services. While fisheries are prominent group affected by new OSW installations, socio-political science also points to a lack of **community consultation around site planning** [23, 1]. Glasson et al. [1] show that EIAs and resulting environmental statements in the past focused on **biophysical impacts** (birds, bats, fish, marine mammals), but hardly on socio-economic ones on local and regional communities, i.e. social, health and economic impacts, and the opportunities that can be created for coastal communities. These may be substantial towards the **regeneration of communities** that suffer from a decline in fisheries, shipbuilding and tourism, but are on many occasions negative for local communities, especially when these are not considered and involved from the start. The authors propose to consider economic factors such as local employment content, local procedural-supply chain, direct/indirect and induced impacts, GVA and GNP, other local business (fisheries, tourism) standardly as part of the EIA process, in addition to social factors such as demographic, housing, local services (health, emergency services), community cohesion and local quality of life indicators. They highlight the Hornsea case study as a positive example which involved developers working with local authorities and businesses to **support education, training and supply chain initiatives** for the Humberside area. Considering the **view of coastal communities** will ultimately help create tipping points [73] for OSW acceptance and public support, alongside factors such as social learning, information provision, framing, role models and changing expectations – and address potentially different international view points on OSW [74] and its legitimisation narrative [22] in terms of sustainability and energy security [75]. Beyond site planning to avoid conflicts, Salvador and Ribereiro [5] offer a view on marine spatial planning from a **legal and regulatory perspective**, studying how the law of the sea can address potential international conflicts between ocean energy installations and other activities (e.g., navigation) developed by other states. This research is in line with other work on the **energy economics** of renewable energy [76], as well as studies that quantity future costs of OSW given different climate models and control variables [77], or work on grid integration models [78] for different turbines models. Other authors point out that **geopolitical contexts** can impact on OSW decisions [79].

Ecological impacts The Mediterranean has been cited as having OSW potential [3, 80], yet some researchers warn that learning from OSW projects in the North Sea may not directly transfer to ecosystems in the Mediterranean Sea. Lloret et al. [7] point to **serious environmental risks** to the seabed and biodiversity from bottom-fixed OSW. They argue that due to the geography of the Mediterranean Sea, most OSW farms would need to be close to shore, creating conflicts for existing biodiversity. The authors further point out that “[t]he overall estimated potential for offshore wind energy in Mediterranean countries such as Croatia, Cyprus, France, Greece, Italy, Malta, Slovenia and Spain for the period 2030–2050 is approximately three times lower than the estimated values of northern European countries such as the United Kingdom, Norway, Sweden, Netherlands, and Denmark (EEA, 2009; European Commission, 2020c)” [80] creating risks that consequences may outweight benefits. Lloret et al. [7] is in line with studies from the North Sea that have shown **positive ecological effects** from OSW mostly from the formation of **artificial reef structures and homogeneous sea beds** [81], which are rare in the Mediterranean Sea and therefore may not transfer. Consequently, multiple studies [82, 83] highlight problems with the idea that offshore wind farms may contribute to biodiversity targets if constructed within Marine Protected Areas (MPAs), as proposed by some authors [84, 85]. Instead, it is proposed that OSW farm should primarily be built in areas that are already industrialised or degraded to prioritise the protection of biodiversity and fish stocks, including the protection of Natura 2000 sites and their buffers zones. This seems crucial given the urgent lack of evidence of long-term environmental impacts of OSW on ocean health. Other other authors are more positive about OSW in Southern Europe, including the Mediterranean Sea [3] and the Canary Islands [86, 8]. Abramic et al. [86] determine the best sites in the Canary Islands based on oceanographic potential, environmental sensibility, restrictions related to marine conservation, land–sea interactions, and avoiding potential conflict with current maritime and coastal activities. They find that **floating wind is particularly attractive** due to steep and deep waters in the Atlantic. However, Rezaei et al. [81] show substantial data and research gaps, especially on the **cumulative effects from larger wind farms**, as current research from historical data shows mostly local and minimal impacts. The authors’ study seems a case in point that observations from foundation-based turbines will not be directly applicable to floating wind. The problem of **substantial data and research gaps** is also highlighted by Watson et al [18] who show that 86% of OSW impacts on ecosystem services are still unknown due to missing research, data or evidence. There are particular risks for the **decommissioning stages** in terms of **habitat destruction** from the removal of artificial reefs that have formed on foundations, and risk from changes to **sediment movements**, mobilisation of **contaminants, fluidisation of seabed and increased turbidity** [20], as observed in earlier work [87]. Gill et al. [88] summarise very recently that there is not currently enough strong evidence to conclude any direct impact of OSW on fisheries species, though the authors warn that indirect evidence should not be discarded prematurely. Guşatu et al. [89] model cumulative impact from various installation stages in the North Sea between 1999 and 2050. They find high impacts in 2022 as a a result of construction caused by **underwater noise, habitat loss, and risk of contact with fuel or chemicals**. Underwater noise is one of the highest pressures and continues throughout operation, while other impacts decrease with the operational phase.

Life cycle There is currently limited published research in the area of life cycle assessment for OSW farms with both fixed-bottom and floating structures. Multiple studies point to specific **data and knowledge gaps** around the impacts of OSW on the environment and natural resources, such as steel as well as critical minerals used in parts of the turbines (e.g. rare earths) [7]. Shadman et al. [91] review environmental impacts from all life stages of OSW in a case study from Brazil, also drawing a comparison with data from the oil and gas sector. The authors find **seasonality** an important factor to mitigate impacts on migrant birds, marine mammals, or sharks and recommend **delaying construction or commissioning** in order to avoid disturbing migration, reproduction, or nesting of vulnerable species. They also highlight that **rich biodiversity** is found in water depths up to 50m, which is suited to bottom-fixed turbines, yet often creates more conflict with ecosystems. This point was also made for the Mediterranean Sea [83], where Brussa et al. find that the most substantial impacts from floating wind stem from the **supply of the floating system itself**, i.e. the semi-submersible platform and mooring chains, with steel being most impacting material [3]. The **high ecological footprint of steel** is further confirmed in Yuan et al. [21] for components manufacturing and transport, wind farm construction, O&M and decommissioning activity for a floating wind farm in China.

Li et al. [92] investigate the environmental impacts of the OSW life cycle under different scenarios until 2040 and find that substantial emissions are created during steel and fibres manufacturing, which the authors argue should **motivate recycling** upon decommissioning. Other factors considered include estimation of OSW

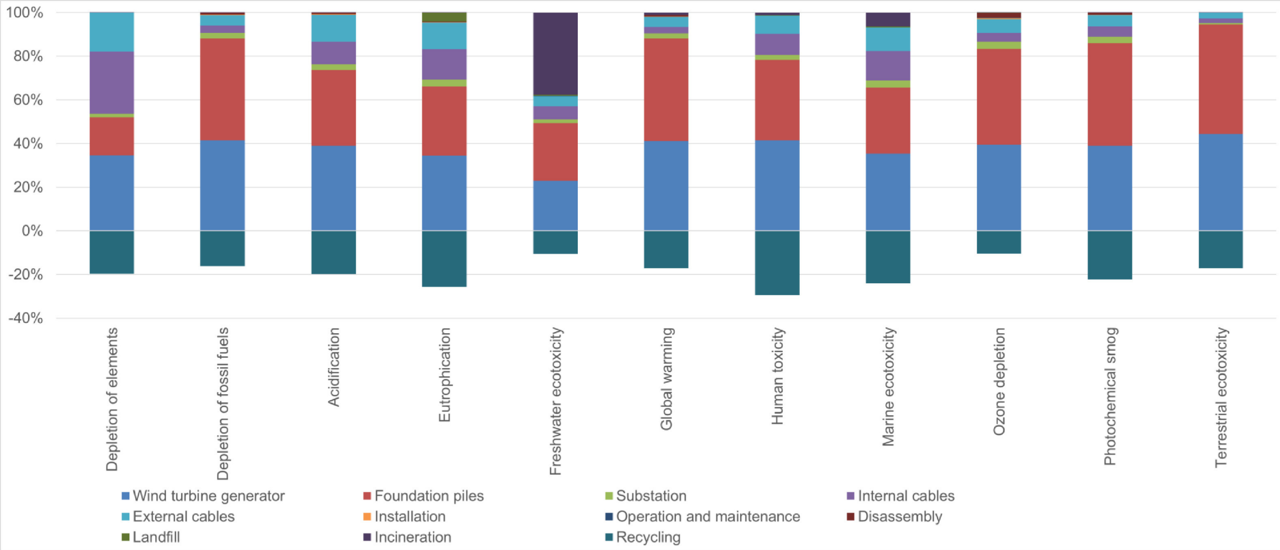

Figure 1: Average contributions of OSW lifecycle stages of the 20GW of UK-installed OSW energy (from [90]), showing that nearly all impacts come from manufacturing, especially due to the large amounts of metal needed for turbines and foundations. Turbines mainly drive pollution affecting people and water, while foundations are the biggest source of greenhouse gases and toxic emissions to land. Cables also contribute to some impacts, like resource use and toxicity. Substations, despite being large, have a relatively small environmental footprint.

electricity generation and life cycle inventory, with overall results pointing to **substantial uncertainty**. This research is complemented by Garcia et al. [93], who estimate the environmental impact for a floating wind turbine during trips and marine operations requires for O&M. Kouloumpis et al. [90] offer a user-centric tool to manage input characteristics of new or existing OSW installation in order to estimate its lifecycle environmental impacts. The authors consider capacity, installation site, wind availability, maintenance requirements and raw materials at different life cycle stages, including landfill and depletion. The study contributes to the narrative on important data gaps on impacts on **marine ecology and atmospheric physical characteristics** from floating wind [91, 3, 93, 21, 92]. This includes Rueda et al. [6] who study the impacts of different materials used in OSW construction, O&M and decommissioning on **human health**, environment and resources. Recent studies have also highlighted the pollution from offshore wind farms in terms of heavy metals and microplastics [94, 95, 96, 97], which requires urgent further research that can in part be informed by data-driven analysis and interdisciplinary collaboration.

Technology Previous research has stressed the growing role of **AI and automation, including robotics**, for the OSW sector [36]. In this context, Mitchell et al. [98] explore in detail and the role Robotics and AI (RAI) can play in the surveying, planning, design, logistics, operational support, training and decommissioning of OSW farms. The authors also offer an evaluation of investment, regulation and skills required to support RAI in the sector, highlighting current barriers of adoption as: certification of autonomous platforms for safety compliance and reliability, the need for digital architectures for scalability in autonomous fleets, adaptive mission planning for resilient resident operations and a human-machine collaboration perspective towards trusted partnerships. RAI is highlighted as beneficial due to the potential of reducing O&M costs, removing people from hazardous environments and situations, as well as addressing personnel shortages, and upscaling the lifecycle process, e.g. via employing robots in installation, decommissioning and logistics. It should be noted that **the authors' perspective** on the use of robots and autonomous systems **is optimistic** in their review paper. While these kinds of systems can be readily deployed for **data collection**, including sub-sea robots for **seafloor inspection** and **environmental impact monitoring**, the state of the art is currently far off being able to offer systems that can autonomously carry out complex repairs or bespoke maintenance operations. Zhang et al. [99] further argue that **UAVs for turbine inspection** can be useful for more remote floating wind structures. Chatterjee and Dethlefs [13] give a comprehensive review of the use and uptake of AI in OSW, particularly for O&M, and highlight **model transparency**, e.g. in neural networks, **lack of high-quality datasets** and performance benchmarks, and a **lack of real-time decision models** as critical challenges. Multiple studies have explored the use of AI and data analysis for **fault prediction, data analysis** and **improved O&M** [100, 101, 102, 103, 104, 105, 106]. For example, Li et al. [34] propose a novel method for O&M optimisation through a divide-and-conquer approach prioritised by operators based on the turbine state. The authors predict that such a closed-loop maintenance strategy can lead to improved profits and revenues. Complementary research has explored the problem of **structural health monitoring** (SHM) [107, 108, 109, 110, 30], such as detecting and identifying damage (or ice) from blades [31], **physics-informed machine learning** to address data sparsity [33], and interactive learning to acquire labels directly from human experts [111]. Related to challenges in transparency and decision support, some studies have investigated the use of **Natural Language Processing** (NLP) and **Large Language Models** (LLMs) for OSW, e.g. for alarm code summarisation [112, 113] or interactive decision support [114]. Bi et al. [115] present an LLM dedicated to the language of ocean science, which fine-tunes a LLaMA-2 model [116] using a dataset of scientific articles on ocean science, engineering and environment.

A small number of datasets exist to encourage **open research and benchmark comparison** in the OSW sector, though these mostly focus on open data, such as remote sensing and satellite images, to reveal the **global distribution** and **installation time** of offshore wind farms [117] and to inform **remote environmental impact assessment** [118]. For example, Hoeser et al. [119] offer a spatiotemporal dataset of Sentinel 1 collected earth observation data capturing just under 10,000 turbines (and substations) at different deployment stages between 2016-21. Other authors use SAR images to compare **wind speeds and wake effects** before and after OSW installations [120], to create future predictions from satellite data for **site planning or energy yield** estimation [121, 122] or to assess **sea state and wave heights** that may affect OSW operations [123]. Most of these studies compare remote sensing data with historical data to gauge the accuracy of future predictions. A key motivation of research on remote sensing is to monitor and assess ecological impacts. Brandao et al. [124] describe **possible effects of sediment composition** around wind farms and present an approach to detect such patterns from satellite images. They are specifically interested in the spatial and temporal patterns and their impact on water quality. Danovaro et al. [125] set out a set of recommendations around siting and monitoring from a study of potential interactions of floating structures with vulnerable marine ecosystems and **critical habitats**, **migration routes** of marine vertebrates and birds and other human interference. A recently emerged concept in OSW is **Digital Twins**, see e.g. [126] for a discussion of definitions. Schneider et al. [127] present a digital twin for monitoring OSW impacts on the ocean from a multi-source dataset of satellite data, local in-situ data and hydrodynamic simulation, to reveal changes in locations, density and temporal variability.

Multiple earlier studies have concluded that currently there is no strong evidence for direct or indirect impacts of OSW on their environment and ecosystems [128, 88]. However this also raises questions about **data collection, monitoring and sensors** and making sure that future OSW farms and environments are equipped with the correct sensors from the get-go. This point is somewhat confirmed in Cresci et al. [129] who highlight that generic environmental monitoring is not sufficient to advise the planning and construction of OSW as it can be unclear what causes certain observations. Instead, the authors propose to use hypothesis testing to determine some of the impacts of new installations in combination with more targeted monitoring. Complementary to studies on new turbine technologies, Akhtar et al. [130] present evidence that **larger turbines** (e.g. 15MW) will lead to **environmental benefits** over multiple smaller structures (e.g. 3 5MW turbines). Jung et al. [40] show that new OSW technologies will also have a greater impact on future wind energy yield than climate change effects, thereby playing down studies that have shown pessimistic future wind forecasts.

One of the most important areas of active engineering research is the design and development of new, larger **floating turbines structures** that can be deployed much further out at sea and in deeper waters, offering greater flexibility, scalability, **increased energy yield** and **reduced social disturbance** and **environmental impact** compared to onshore installations, at least from bottom-fixed structures. It has been noted that an overall lower environmental impact may not apply, and floating technologies instead come with different impacts, such as the effects of anchors and chains, which need researching and documenting. Hong et al. [4] discuss technical, operational and economic opportunities and challenges. The authors discuss opportunities and highlight a multitude of **open research questions** around **installations** (cables, moorings, substations), **cost, planning, fabrication, logistics, maintenance, resources**, etc. Besides, there are challenges around the industrialisation of areas with OSW (if not industrialised before) if these are within or close to MPAs – as OSW will then typically entail the industrialisation of the full MPA, which is problematic [131]. In Europe, France is currently the only country with floating wind while the UK is expected to have over 20GW of floating wind turbines installed by 2032 [132], while the global target for 2050 is 270GW capacity installed floating wind. Current research and planning focuses on areas in the North Sea, the Baltic Sea and the Galician Coast [4], while other authors have explored potential in the Mediterranean Sea [7, 3] and the Atlantic [86]. Considering the levelised cost of energy (**LCoE**), Hong et al. state that **costs will vary** depending on the type of floating foundation, installation process, and distance from shore, yet in general it increases proportionally with the distance between wind farm and shore and the use of installation vessel. Specific challenges are the requirements for **fleet expansion** of installation vessels, **vessel upgrades** to service larger structures, as well as new vessel designs to improve efficiency and capability. Beyond capacity limitations of the current OSW sector, the authors highlight research gaps around **inadequate weather planning** and impacts, **foundation and anchor design**, e.g. in relation to **seabed interactions**. They further highlight the need for more **stability analysis** of floating foundations given different environmental conditions, as well as during the towing and installation process. Finally, they stress the potential of **digital twins** for predicting structural responses.

While increasing autonomy offers benefits for O&M, SHM and others, it also presents new challenges relating to potential cyber attacks. Kam et al. [44] estimate that cyber attacks can cause operational shutdowns (85%), damage to energy assets and critical infrastructure (84%). These statistics are based on case studies from Germany (distributed denial of service attack on email), the US (firewall reboots) and multiple others involving communication network breakdowns. The authors highlight that **operational and information technology increasingly interlinked**, creating urgency for improved cyber-resilience measures in the OSW sector. Hoppa [45] studies the risk of cyberattacks on OSW infrastructure, mapping out realistic scenarios for **physical access attacks** at wind turbine or substations, as well as **cyber access** via vendor networks, technician equipment or compromised supply chains. The research highlights known vulnerabilities such as turbines with direct public internet connections or legacy software, such as Windows 2000, which may enable information disclosure, malware injection or denial of service attacks. The study concludes that the current risk of cyber is low, but that risks from **criminals, nation state actors, insiders, terrorists** or **activists** may grow in future with an expanding sector. Cyber risks can apply to OSW infrastructure or **related logistics**, such as shipps and vessels [46]. Tam et al. [47] consider larger systems of **interconnected energy systems** and identify secure channels, updated safety standards and improved personnel training as immediate mitigations. Cyber-risk to OSW infrastructure will need to be modelled in future alongside existing risks of accidents which can arise from automation (collision with ships, technical issues such as blade failures, fire, storms, etc.).

Workforce & Training In contrast with other sectors that have well-established health and safety frameworks and corresponding training programmes, OSW is still developing in this regard and has been shown to lag behind [9]. Previous research in this space has often focused on a particular sub-aspect of risk. For example, Rawson et al. [133] present a model for **navigational risks** (e.g. around wind farms for other existing sea users) assessment that seeks to directly address shortcomings in previous work that overestimate navigational risk due to not taking historical data into account. The authors present a comprehensive alternative model including factors such as **collisions** with wind farms, resulting **environmental pollution** and **injuries**, but also **ferries and good transport** that is pushed to other shipping routes, increasing transit time, impacts from wind and tidal changes as well as impacts on radar coverage. The authors argue that navigational risk should be modelled as part of the environmental impact assessment. In a complementary study, Lin et al. [134] focus on risk assessment and management for **activities in ports**, e.g. that support OSW operations without directly taking place offshore. The authors work from a mixed-methods study including interviews, on-site visits and observations. Earle et al. [135] study the **risk on offshore operations from the physical and mental state** of technicians working offshore, e.g. impacted by sea sickness and other factors stemming from vessel transfer or spending extended periods of time offshore. Also focusing on human behaviour offshore, Wang et al. [24] offer an analysis of **human factors that lead to accidents** on offshore vessels, e.g. during site development or maintenance operations. The authors analysed past accident scenarios and find that 60-90% of accidents in their scenarios are ultimately related to human factors, giving rise to **increased needs for training**.

The most comprehensive health & safety review for OSW is Rowell et al. [9], who reviewed the state of the art in safety in OSW, also considering the **state of legislation and injury statistics**. Their study finds that OSW operations are currently **3-4 times more risky** than any other sector in terms of injury statistics, and accidents are substantially more likely to occur in OSW than related sectors, such as oil and gas. They further find that there are currently **no standard official reporting procedures** for OSW accidents in the EU, leaving a substantial gap. The authors derive their results from analysing total incident rates, occupational illness frequency, fatal incident rate, recordable injury rate, amongst others, which they relate directly to **technical failure**, **work environment** and **training**, **transport and traffic** as well as a number of **external factors**. The review concludes that with a rapidly growing workforce (which is projected to triple in the next five years), new challenges such as **remote worksites**, **accessibility** issues and **lower profit margins** all act as risks to H&S performance in the sector. This may be exacerbates by **supply chain pressures**, such as the increasing use of subcontractors due to skills shortages. While **increased automation** was highlighted as an opportunity, it comes hand in hand with new risks around human-AI collaboration and **cybersecurity** risks discussed above. Alarmingly, the authors conclude that green objectives that drive a rapid OSW expansion seem to act as a direct motivator of reduced occupational health and safety. Roberts & Flin [10] find lagging health & safety in a recent data analysis study confirming the risks of working in the sector. They report that while 2023 saw a 39% increase in the total number of hours worked over 2022, the recorded incident rate increased by as much as 94%. The authors devise a framework that distinguishes health and safety hazards across four domains: individual (e.g. fatigue, decision-making), team (e.g. communication, situational awareness), task/environment (e.g. confined spaces, weather, noise), and organisational factors (e.g. safety leadership, culture, training). The study concludes a **general lack of research** on the psychological and organisational factors that impact on wind technician safety, health, and performance, confirming a major research gap.

C.2 Progress Beyond State of the Art

Current research highlights numerous **gaps in knowledge, data, benchmarks, practice and regulatory frameworks** that present significant barriers to the rapid expansion and innovation that is required to meet 2030 (and 2050) net zero goals. Offshore wind is central to these net zero goals as well as to achieving **global sustainable development goals** for affordable and clean energy (SDG 87), decent work and economic growth (SDG 8), sustainable cities and communities (SDB 11), climate action (SDG 13) and life below water (SDG 14).

1. The sector still lacks comprehensive frameworks that **consolidate state-of-the-art knowledge and lay out roadmaps** for innovation.
2. There are persistent gaps in understanding the **ecological impacts of OSW**, particularly concerning decommissioning and the transferability of insights from mature sites like the North Sea to new regions, such as the Mediterranean.
3. Socioeconomic trade-offs, including site planning conflicts, **multi-use optimisation**, and standardised **integration of social benefits** into impact assessments, remain underexplored.
4. Research into **advanced materials**, especially high-performance steel needs for floating wind, and pathways for **sustainable recycling** is extremely limited.
5. Technological gaps include **trustworthy, transparent model** that handle **sparse data**, via either new research into efficient learning algorithms or new frameworks for data sharing.
6. Research and technological innovation is needed on **robust, human-in-the-loop approaches** for **robot-augmented operations** and UAV-assisted **inspection and monitoring**, including **data-efficient neuromorphic learning models** that can learn on devices, and strategies to **build cyber-resilience** to protect the increasingly automated OSW infrastructure.
7. There is a pressing need for **pre-planned sensor networks**, **long-term data collection** frameworks, new **digital twin** applications, and **targeted workforce development** initiatives, including **standardised European-wide health and safety protocols** and reporting frameworks.

D RATIONALE FOR CHOOSING NETWORKING TO ADDRESS THE MAIN CHALLENGE

This Action views networking as the most effective mechanism to support a safe, just, sustainable, resilient, and digitally-enabled expansion of offshore wind (OSW) across Europe. The pace and scale of planned OSW development over the next five years demand a response that extends beyond isolated research or national-level strategies. Networking enables us to coordinate knowledge, align efforts, and accelerate innovation across geographical, disciplinary, and institutional boundaries. It offers a framework for addressing shared challenges collectively, rather than duplicating efforts or reinventing solutions in national silos. The European OSW landscape is currently unbalanced, as illustrated in Figure 2. While some countries lead in deployment and innovation, others are only beginning to explore their offshore potential, see also Europe's own map of wind capacity. Networking creates opportunities for meaningful knowledge transfer, but it also ensures this exchange is not one-directional. Instead, the Action prioritises inclusive engagement, where local needs and expertise shape how solutions are adapted to regional contexts. Bottom-up mechanisms such as community focus groups and regional and national workshops with international participation ensure that the views and experiences of coastal communities, sea users and diverse stakeholders are integrated into policy, practice and education.

This COST Action has potential beyond single European, e.g Horizon Europe, projects, to engage a much larger network of partners and participants to create strong momentum beyond single projects. This is even more urgent given the fact that much about the impact of OSW is not yet known, so the mission of this Action to discover data and research gaps, in collaboration and communication with all European countries, is important and addressed through networking. Networking is also key to addressing systemic barriers in the OSW ecosystem. These include limited cross-national collaboration between researchers and stakeholders, inadequate access to cutting-edge research facilities (especially in digital and ICT domains and especially for researchers from

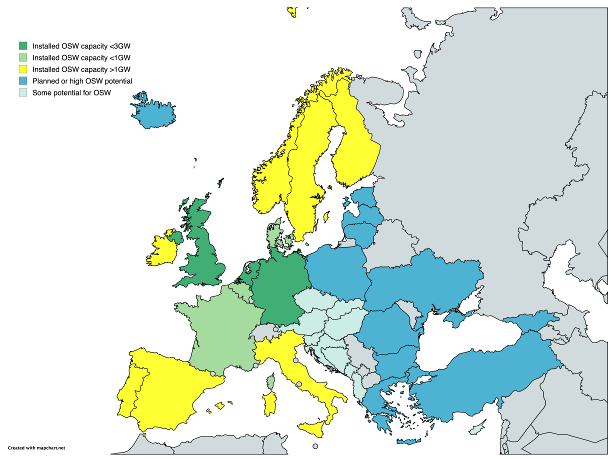

Figure 2: Map of installed and planned offshore wind capacity in Europe including countries with OSW potential. All coloured countries are involved in this Action (including the Canary Islands which are not shown) with

Inclusive Target Counrries), and disparities in workforce development. By enabling mobility and collaboration, particularly for ECRs and underrepresented groups, the Action supports a more equitable talent pipeline and helps reduce gender and regional disparities in the sector. Through shared infrastructure, coordinated training, and mentorship, it directly contributes to building capacity for a sustainable OSW sector. Achieving the proposed research and capacity-building objectives outlined above is only possible through funded collaboration across a wide-ranging, multidisciplinary, and international network– which does not currently exist in Europe or globally. Without a network, we risk a situation where knowledge remains in certain countries (those that lead on OSW, see Figure **??**) or is transferred to new projects without relevant knowledge of the context of the target country – see e.g. [83, 7, 39] as a caution of directly transferring knowledge from OSW in the North Sea to the Mediterranean. The distinctive strength of this Action lies in its ability to bring together experts, researchers, innovators, policy makers around shared goals, enabling coordinated data collection, comparative analysis, knowledge sharing, and more effective translation of theory into practice.

As the first COST Action that is explicitly dedicated to OSW, this network provides a unique platform to co-create a coherent, pan-European research and innovation agenda. By identifying synergies, aligning priorities, and building lasting relationships, the network will not only support current deployment efforts but also lay the groundwork for long-term resilience, adaptability, and excellence in OSW. Crucially, it ensures that inclusion across geography, discipline, gender, and career stage is built into the foundation of this transition, making it socially as well as technologically sustainable.

E CRITICAL MASS OF THE NETWORK

This Action links zzz stakeholders across xx countries, yy disciplines, and zz sectors, bringing together expertise on environmental science, engineering, policy, digital infrastructure, and social impact. It thereby fosters a shared understanding of the risks and opportunities associated with OSW and promotes integrated, systems-level solutions that no single institution or country could achieve alone. Current members of this Action involve all 27 member states [IF THIS IS THE CASE] and full COST members, achieving a balanced representation with Inclusive Target Countries. The network also includes members from Outer Regions with strong OSW potential. MENTION USA.

Over the course of this Action we aim to build a large and active pan-European community that also involves active collaboration with near neighbour countries and other global players in OSW. We want to create a bottom-up community focused on the safe expansion of offshore wind, ensuring benefits for coastal communities, industries, and ecosystems. Inspired by the Climate Change AI community, which has 13K+ members, we will create a strong network with open membership, regular newsletters, geographically distributed research workshops, funding opportunities e.g. for travel, meet organisation, doctoral co-supervision, and as seed priming funds for new research collaborations. The Action will create an active blog with updates from the community, authored by different members or associations, organise hackathons, summer/ training schools, and jointly author policy reports. To achieve this, the Action will develop a website showcasing its mission, key themes, partners, and regional priorities. Social media channels and a LinkedIn group will foster engagement and grow its digital presence. A wiki/data repository will host scientific articles, public datasets, policies, and regulations, inviting contributions to expand knowledge sharing. The Action will organise international symposia, workshops, and satellite events at major conferences, featuring invited talks, panel discussions, and research presentations. These efforts will raise awareness, attract members, and drive impactful collaboration in the OSW sector.

2 IMPACT

Our end goal is a self-sustained international expert network that is future-proof and open to new members. We consider short-term and long-term impact relating to science and technology, economic, social, environmental and political impact, implemented through an inter-, multi-, and transdisciplinary perspective that enables research and innovation through new knowledge, access to infrastructure e.g. research facilities and expertise.

A IMPACT RELATED TO OBJECTIVES

- **Knowledge**: structured accumulation, consolidation and documentation of: (A) research, scientific evidence, data and evidence gaps (reports, article, white paper); (B) open access sources of information and data (wiki, data repository) for international knowledge sharing and exchange between different types of stakeholders; (C) existing frameworks, best practice. The Action aims to bridge disciplines to create new knowledge that is more than the sum of its parts. This relates to **Research Coordination** objectives 1-5 (Sec. B.1) and is achieved through WGs X-Y.

- **Collaboration**: facilitated joint research projects, grand bids through network; improved access to knowledge, data and research facilities internationally leading to more impactful research; cross-border pilot projects and initiatives via network pump priming; joint publications, exchange programmes; stimulate innovation and increase representation in the OSW sector. This relates to **Capacity Building** objectives (Sec. B.2) on building an expert network, creating research clusters/consortia, collating/curating new datasets and will be implemented via WGs X-Y around focus topics.
- **Education**: documented need for new training programmes (offshore and port operations, safety and risk management, environmental impact assessment, AI, data analysis and digitalisation) at different levels (professional, higher education, schools); established mentorship programmes via expert network; Short-term scientific missions; support of Early Career Researchers; webinars, shared educational materials, delivery of educational programmes + hackathons; career development and networking for ECRs. This relates to **Capacity Building** objectives (B.2) on creating cross-disciplinary education (and research collaboration) networks and addressing under-representation and intersectionality through inclusive education and training. It addresses **Research Coordination** objectives (B.1) by creating new knowledge to inform sector-specific education and training programmes.
- **Policy**: documentation (white paper/s) of regional and global needs, regulatory barriers and gaps; roadmaps for shared standards, safety frameworks and protocols taking into account ethical issues; increased influence on policymakers, guide citizens and educators, evidence-based policies through collective expertise. Cf. **Capacity Building** (B.2) on expert network, data curation; **Research Coordination** (B.1) on streamlining regulatory processes.
- **Visibility**: high-impact publications in premier outlets, such as Nature Communications, and public-facing publications on pressing sector issues; raise public awareness (citizens, media, diverse stakeholders, think tanks) to stimulate and support dialogue; established and growing online community via website, social media channels. Emerging from **Capacity Building** (B.2) on active dissemination, citizen and stakeholder engagement, implemented through a dedicated WG X.

Short-term (measurable) impacts In the short term, the Action will generate structured knowledge resources, promote data sharing and standardisation and strengthen collaboration across research groups. Early outputs, such as joint publications or datasets, as well as pilot projects, e.g. via research visits and Short-term Scientific Missions, will accelerate new research and innovation and contribute to advancing impact in science and technology. At the same time, increased visibility and public engagement will raise awareness of the offshore wind sector and its role towards carbon neutrality, thereby promoting careers and study pathways towards building a talent pipeline. Mentorship programmes and exchange opportunities will directly benefit early career researchers, building capacity across diverse communities, especially combined with inclusive education and training initiatives.

Long-term (measurable) impacts In the longer term, coordinated research clusters and consortia, international knowledge exchange and collaboration will establish common standards, frameworks, and interoperable datasets. These will enhance the reliability of research outcomes and support innovation in areas such as engineering and digitalisation, risk management, and environmental sustainability, while regulatory roadmaps will reduce barriers to deployment and ensure safety and sustainability. Similarly, the Action will generate stronger international policy influence, evidence-based regulation and shared European standards. By embedding inclusive education and cross-border collaboration, it will support a resilient self-sustained ecosystem of experts at different career stages, capable of addressing future challenges in the offshore wind sector.

B INVOLVEMENT OF STAKEHOLDERS

There are four main groups of stakeholders that this Action aims to communicate and collaborate with: (1) **Industry and end users**, such as wind farm owners/operators, and developers and related specialised industries that serve the sector (e.g. blade manufacturing, sensor production, logistics, shipping), (2) **policy makers and regulators**, such as governments (from local councils to national governments), NGOs, think tanks, port authorities, managers of marine protected areas, etc. (3) **community groups and citizens**, including schools, professional and recreational fishers and local tourism businesses, and (4) the **scientific community**, including researchers and educators to support the growing OSW sector. The Action will work with a bespoke **Stakeholder Outreach and Engagement Strategy** to establish and maintain communication channels and seek stakeholders views to directly impact planned activities and meetings. This will be developed as a priority Deliverable in Month XXX, see Section **??**. Stakeholders will be invited to the following activities:

- Attend the Action's kick-off meeting and satellite consultative workshop to help define immediate and long-term priorities for the network.
- Join and participate in the Action's stakeholder advisory board (held six-monthly) to receive updates from the MC and WG leads, and to provide feedback on Action direction and priorities for the next six months.
- Attend the 2-day XXX Action conference in Years 2 and 4, featuring presentations and discussions on key network achievements from the preceding 24 months.
- Participate in WG meetings, organised bi-weekly by WG leads, and attended by cross-country and cross-disciplinary members.
- Host researchers (with a particular focus on ECRs) with stakeholders or local communities for short stays to support targeted collaboration, consultancy, and/or facility access to faciliate OSW innovation.
- Participate in workshops (held six-monthly in different partner countries and regions) around specific OSW topic areas defined by WGs and the stakeholder board.
- Sign up for weekly newsletters to receive updates on Action activities, meetings, publications, and collaboration opportunities.
- Participate in our expert web forum to initiate or contribute to virtual discussions on sub-topics in OSW.
- Join our virtual expert network via the dedicated website portal and/or LinkedIn group.
- Engage in data sharing, hosting, and standardisation initiatives organised through our DATA WG.
- Participate in our ECR mentoring programme, supporting the next generation of offshore wind researchers and practitioners.
- Propose, review, and collaborate on proof-of-concept projects and pump-priming activities for short-term missions and projects.
- Host or co-fund technical meetings such as summer/winter schools or hackathons around OSW topics defined by WGs and the stakeholder board.
- Feed into our education programmes to provide industry and third-sector perspectives on training and skills gaps in offshore wind, ensuring programmes reflect sector needs.

These engagement activities are indicative and will be concretised and expanded within the Stakeholder Outreach and Engagement Strategy.

C COMMUNICATION, DISSEMINATION AND VALORISATION

We following activities are planned to maximise communication, dissemination and valorisation:

- Organisation of targeted workshops and roundtable discussions, where events are held six-monthly in different countries/regions and focus on specific sub-topics, informed by WGs, to implement a bottom-up approach for network participation. Events involve researchers, industry, policymakers, and citizen groups (including local communities affected by OSW development) and focus on knowledge exchange and input collation for roadmaps and white papers.
- Hosting and maintenance of a dedicated network website to act as a primary hub for information dissemination on the network and members, announcements of events and news, blog posts, hosting of resources, publications and links to data, code repositories and educational materials. The website will provide an easy mechanism for new members to join the network and/or mailing list and feature an interactive discussion forum.
- Regular (weekly) newsletters will communicate updates via a mailing list on research outputs and publications, events, funding opportunities (e.g. for research exchanges, industry placement opportunities, research facilities, etc.) and collaboration opportunities.

- Publicly accessible repositories will host datasets, code and tools, protocols, and project deliverables, ensuring transparency, reproducibility, and knowledge valorisation. Standards for data hosting and sharing will be defined within the DATA WG and follow established standard such as A, B, C.
- The Action will organise Special Issues in high-visibility journals on inter-disciplinary topics related to OSW as part of the overal dissemination strategy (defined in XXXX WG). WG will define publication plans that factor in special issues in journal such as X, Y, Z.
- Development of a flagship “Action book” consolidating key findings, best practices, and policy recommendations. This will serve as a long-term reference for the offshore wind research, industry, and policy community.
- Co-production of teaching materials at different levels, including schools, colleges and universities will support the creation of a talent pipeline to attract more people to work in the OSW sector. WGs will develop materials to support university modules, training curricula, and case studies for higher education, professional training, and schools. Materials will be openly available and disseminated through partner institutions and the Action website.
- Participation in science festivals, citizen group consultations, and community-facing events, including and tourism and fisheries events, school visits, will help to broaden public awareness, give opportunities to discuss the role of OSW in supporting Europe’s energy systems and offer opportunities for hands-on activities such as science experiments. To embed inclusivity, some activities will be tailored to underrepresented groups to maximise societal impact, the details of which need to be developed in collaboration with the Action’s dedicated EDI LEAD. Public events like the above will also be developed with regional representatives to ensures suitability for different geographical and sociocultural areas.
- Social media channels (LinkedIn, BlueSky, ResearchGate, etc.) will help communicate network results widely, encourage discussion, and maintain visibility beyond the project, and act in parallel to the project website. A set of central social media accounts led by the network’s dissemination LEAD will be supported with activity on individual member accounts.
- A dedicated Dissemination Coordinator will be appointed to lead on Action dissemination and coordinate communication streams from individual WGs. This involves ensuring that dissemination activities target the diverse audiences in the right way and aligning them with the project’s exploitation plan.
- A detailed exploitation plan will be developed to outline how research outputs (datasets, methods, policy recommendations) can be implemented by industry, policymakers, local NGOs and public sector, ensuring long-term valorisation of project results. This can also inform future calls for research projects to investigate in-depth areas of knowledge gaps.

3 IMPLEMENTATION

working groups and a management committee; each WG has a focus, their output needs to be collated, specify membership and probably show a diagram; this can be similar to Aura

also a steering committee; people more peripheral such as larger non-academic entities

All WGs collate ongoing lit review and SOA of their theme, discuss newest trends, document, share expertise, set out needs, distribute updates, funding opportunities to spin out projects from reviewing and roadmapping activities.

One WP on impact and policy, de-risking technology

A ACTION STRUCTURE

B WORK PLAN (TASKS, ACTIVITIES AND TIMEFRAME)

C DELIVERABLES

D GANTT CHART

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